Reet Mishra

Machine Learning · Data Science · Design

EMAIL choconlily@gmail.com
PORTFOLIO ciiphersiren.github.io
GITHUB github.com/ciiphersiren
LINKEDIN linkedin.com/in/ciiphersiren

PROJECTS

what are you made of? | *real-time webcam piece* 10.2026

JavaScript · Canvas · MediaPipe · WebAssembly

- Redraws a person's live silhouette in the words they typed at **20–50 fps**, fully on-device: face and body segmentation via WebAssembly, thousands of tinted glyphs per frame on the GPU
- Grapheme-aware text so any script or emoji fills one cell; a background model learns the room and paints it back to make the user disappear

Recomme To Me | *recommender system* 09.2026

Python · NumPy · Flask · scikit-learn · JavaScript

- Implemented 5 recommenders from scratch (item CF, TF-IDF, SGD matrix factorization, Bayesian popularity, hybrid) behind one interface and 12 Flask REST endpoints
- Designed a leakage-free evaluation harness on MovieLens (per-user temporal split): RMSE, Precision@10, recall, coverage, diversity, novelty

matlab-ascii-art | *image processing in base MATLAB* 10.2026

MATLAB · matlab.unittest · GitHub Actions

- Developed an image-to-ASCII converter with color output, TXT/PNG/HTML export and an app; reimplemented resizing, grayscale, equalization and sharpening without toolboxes
- Vectorized block averaging (up to 6.2× faster than a loop) and benchmarked where it turns memory-bound; **80 tests, 97.9% coverage** behind a CI gate; on File Exchange

learn-to-RL | *interactive RL visualizer*

JavaScript · Canvas · zero dependencies

- Created a browser grid world where a Q-learning agent learns in real time, with Q-values drawn as arrows and live reward charts

Prompt-Based Object Detection | *team capstone* 08.2026 – Present

PyTorch · YOLO26 · YOLOE · SAHI

- Trained YOLO26m on VisDrone aerial imagery with ignore-region masking and multi-scale tiling, reaching **0.636 AP50** on val with SAHI; adding YOLOE for open-vocabulary prompts, with an IEEE paper in progress

RESEARCH

Neuromatch Academy | *project pod · ECoG signal analysis* 07.2026

Python · MNE · FOOOF · statistics · classification

- Led a four-student pod analyzing intracranial (ECoG) recordings from a working-memory task; wrote the team's MNE + FOOOF preprocessing pipeline as primary coder
- Traced an initial null result to a normalization artifact; after fixing it, found the effect in 2 of 3 subjects and **above-chance multiclass decoding in all 3** (permutation $p < .001$)

LEADERSHIP

Vice President | *Open Source Community, VIT-AP* 10.2024 – 03.2026

- Headed an 85+ member community and organized 10+ workshops and hackathons; rose from Events Team to Events Lead, Track Lead, then VP

President | *Madhya Bharat Association, VIT-AP* 08.2024 – 06.2026

- Managed teams of 50+ across design, content and logistics; raised member retention 30% and event attendance 60%; rose from Design Team through Treasurer and VP

Design Team Lead | *Nrityam, Anchoring Club, VIT-AP* 09.2024 – 09.2025

- Directed design for two clubs, delivering visual assets for 5+ events with 500+ attendees
- On GDG on Campus's design team; organizer of **GoogleDevs Sprint 2K25**, a hackathon run with Google Cloud (03.2025)

EDUCATION

VIT-AP University

B.Tech in Computer Science Engineering, specialization in AI/ML
CGPA 8.5 / 10
2023 – 2027

COURSEWORK

ML · Machine Learning, AI, Reinforcement Learning, NLP, Applied Statistics
Core CS · DSA, DAA, DBMS, Operating Systems, COA

FIITJEE

Class XII, PCM · **93%**
2020 – 2022

Vista International School

Class X · **93%**
2016 – 2020

SKILLS

LANGUAGES
Python, Java, C, C++, JavaScript, MATLAB, SQL

MACHINE LEARNING
PyTorch, Ultralytics YOLO, SAHI, scikit-learn, NumPy, MediaPipe, MNE-Python; NLP, RL, recommender systems

BACKEND & WEB
Flask, Node.js, Express, PostgreSQL, REST APIs

TOOLS
Git, GitHub Actions, Claude Code, Jupyter, Figma, Canva

AWARDS

Kaggle contest · 3rd place

NPTEL HCI · Elite + Gold

NPTEL Classics in Neuroscience · Elite + Silver

Udemy · Full Stack Web Development

AI Foundations · Thinking Machines

LANGUAGES

English	fluent
Hindi	fluent
Korean	intermediate
German	elementary
Chinese	elementary